The snowmelt energy balance during the March 2026 heatwave: A case study in the Northern Sierra Nevada

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Abstract
Test

1 Introduction

While meltwater from seasonal snowpacks has historically provided ∼70% of runoff in the western USA (Li et al., 2017), these natural reservoirs and our ability to measure, monitor, and predict them are in decline (Hale et al., 2023; Cowherd et al., 2024). From a practical perspective, many operational snowmelt runoff models use a temperature index approach, whereby air temperature (T_{air}) is the primary predictor of melt (Anderson, 1973). While these models have proven effective (after calibrating empirical coefficients for each site or watershed), the physical justification for such models is not clear. Since at least the 1980s the field of snow hydrology appreciates that understanding the climatic behavior of snow necessitates considering the entire energy budget of the snowpack. In particular, net shortwave radiation incident on the snowpack (SW_{net}) is a first-order predictor of rates of melt, and it is well accepted that both small impurities (Deems et al., 2013) and natural grain-growth mechanisms that decrease snow albedo (α) TODO: cite Warren, Colbeck are responsible, in conjunction with rising sun-angles in the spring, for the majority of energy available for snowmelt. Unlike atmospheric radiation in the longwave portion of the spectra (4-100μm), shortwave (0.3-3μm) is mainly insensitive to variations in T_{air} itself. Variations in $SW\downarrow$ in time (and across elevation) are driven primarily by Rayleigh scattering by O_2 and N_2, Mie scattering by natural occurring and human-made aerosols, dust, and cloud droplets, as well as absorption primarily by water vapor, CO_2, and O_3.

At the same time, recent work has highlighted the strong correlation between extreme snow ablation events and so-called "snow-eater heatwaves" — prolonged periods of sustained (day and night), anomalous, and above freezing T_{air} during the snow season TODO: CITE Rhoades, in press. The characteristics of snow-eater heatwaves have already appreciably changed since the 1850s. These events are occurring more frequently, at a higher intensity, over a larger portion of the western US (WUS), and earlier in the water year.

An example of such an early-season, high-intensity, snow-eater heatwave occurred in March of 2026. This event, which started earlier than the typical window of occurrence studied in TODO: CITE Rhoades, in press, resulted in widespread and rapid melting at many snow-observing sites and one to two months earlier than average streamflow in many basins. This raises an apparent paradox: if $SW\downarrow$ provides most of the energy for snowmelt, and yet it is largely insensitive to T_{air}, then why are heatwaves so efficient at ablating snow?

Using the March 2026 event as a case study, we ask the following questions:

- What are the dominant snowmelt energy forcings during early-season heatwaves, and how do they scale with increasing T_{air}?

- How do vegetation canopies and atmospheric conditions — including cirrus clouds — shape such energetic forcings during heatwave events?
- What are the relative contributions of sublimation and melt to extreme snow ablation during such events?

For some perspective, approximately half of North America's snow water volume resides under tree canopy Kim et al. (2021). Canopies absorbs shortwave radiation, increases the longwave radiation, and reduces winds relative to open or non-vegetated snow surfaces, however the magnitudes of these changes can vary widely across space and time. Though little work has examined how snow responds in vegetated versus non-vegetated regions during heatwave events TODO: but, may knows of a paper that has looked at this.

While low clouds have the most substantive effect on the snow energy balance (Rudisill et al., 2025), despite otherwise drier-than-normal conditions, high altitude cirrus clouds can occur during high-pressure ridging events commonly associated with cool-season heatwaves (Cite Rhoades et al. snow-eater heatwave paper, such as Figure 1 and Supplemental Figure S5). Cirrus increases longwave and decreases shortwave relative to clear skies (CITE), though the radiative impacts to snowpacks are underexplored in the literature, let alone during heatwave events.

To answer these questions posed earlier, we use the March 2026 event as an object lesson to explore heatwave-snow interactions and place the record heat and associated snowmelt in a climatic and synoptic context. We focus our analysis on the greater Northern Sierra Nevada, which supports runoff into the American, Yuba, and Lake Tahoe/Truckee River basins. To do so, we use multiple lines of evidence (reanalyses, long-term station records, remote sensing observations, and models) to infer snowpack characteristic changes (e.g., albedo, snow drought classification, and snow-covered area) and energy balance shifts before, during, and after heatwave events.

The outcomes from this study answer basic science questions about energy available for snowmelt or sublimation in water-resource critical basins in the northern Sierra Nevada, with potential implications for other regions. The findings can directly inform efforts towards management and resilience against extreme events by providing actionable steps towards improving the monitoring and prediction of snowmelt. Last, improved understanding of vegetation and snow interactions during heatwaves, as well as during other runoff-inducing events such as typical warm-day melt and rain-on-snow, can inform headwater management practices through improved decision support tools (e.g., Heggli et al. 2022; 2026).

2 Background and Methods

2.1 Study area

We focus our analysis on three basins in the immediate vicinity of the the Central Sierra Snow Laboratory (CSSL; 39.32 °N, 120.37 °W, 2100 m.a.s.l) located in Soda Springs, CA (Schwartz et al., 2026). These include the Yuba, North Fork American, and Truckee River Watersheds (the greater Truckee River Watershed is composed of the Lake Tahoe basin and Truckee River basin at the eight-digit hydrologic unit code (HUC8) scale) (Fig. 1a–d). The Yuba and North Fork American rivers support the New Bullards Bar and Folsom reservoirs, respectively. These reservoirs have combined storage capacities approaching 2 million acre feet and hydropower generation potential of 559 megawatts TODO: citation. All three rivers support varied downstream consumptive uses (i.e., urban, industrial and agricultural) and provide critical aquatic and riparian habitats. TODO: Statement about how vegetated the main snow regions are and how representative the forest LAI site is.

2.2 Observational and Reanalysis Data

We use data from twelve Natural Resources Conservation Service (NRCS) SNOwpack TELemetry (SNOTEL) Network stations located throughout the study region spanning TODO: X to TODO: Y m.a.s.l. SNOTEL is the primary snow observing program in the western USA (Serreze et al., 1999) and routinely measures snow water equivalent (SWE), snow depth, and air temperature. Homogeneity of air temperature measurements remains a challenge for the SNOTEL network (Oyler et al., 2015; McAfee et al., 2019). To supplement SNOTEL temperature records, we use data from four US Historical Climate Reference Network (USHCRN) stations and the gridded nClimGrid product throughout the study region. USHCRN stations are a subset of the broader GHCN-D dataset Menne et al. (2012). The HCNv2 dataset provides homogenized and quality-assured monthly air temperature dataset based on USHCRN data that dates as far back as the late 1800s (Menne et al., 2009). While this data does not have the same spatial density and elevation range as SNOTEL, the longer temporal record and high quality control mean that these observations can more confidently be used for evaluation of climate extremes, such as heatwave events (Bercos-Hickey et al., 2022).

In addition to hosting an NRCS SNOTEL site, CSSL provides long-term meteorological and snowpack records in addition to surface energy balance measurements. Campbell Scientific CNR4 four-component radiometers measure incoming and outgoing SW and LW radiation from the atmosphere and snowpack. A Campbell Scientific IRGASON eddy covariance system measures sensible (H_s) and latent (H_l) heat fluxes above the snowpack. Both the IRGASON and CNR4 are hosted on a movable arm (Fig. 1c) that automatically adjusts to maintain a constant distance of approximately one meter above the snow surface as snow accumulates or ablates throughout the season. Eddy covariance fluxes were processed using the Licor EddyPro v7.0.9 software with despiking, double coordinate rotation, and flux quality assignment. TODO: X% of the H_s and H_l fluxes are retained for analysis during the month of March.

The CSSL forest study site hosts an additional meteorological station approximately 150 m North of the main CSSL site. This tower hosts an additional Kipp and Zonen SP Lite2 radiometer that measures below-canopy up and downwelling shortwave radiation. SWE measurements were obtained using a federal sampler (Osterhuber, 2014) at nine discrete sampling intervals during the March heatwave.

To investigate the synoptic patterns associated with the March heatwave, we use 500 hPa geopotential height and atmospheric column information from the ERA5 reanalysis (Hersbach et al., 2020). We use the SPIReS (Bair et al., 2021) dataset to investigate changes in snow-covered area and snow α during the heatwave. The SPIReS algorithm is based on observed reflectances by instruments on the MODIS polar orbiting satellite and provides daily data from 2002 to present. As a measure of the canopy temperature, we use data from the VIIRS/Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature (LST) data product for day and nighttime (Hulley and Hook, 2018). The canopy temperature data has a resolution of 1km and provides twice-daily observations of LST.

2.3 SUMMA Model Configuration

To investigate the snow energy balance, we use the SUMMA model (Clark et al., 2015). SUMMA is well-vetted for a range of snow climate applications and provides several options for flux stability corrections, canopy radiation interactions, snow-layering schemes, and other model structural options (Cristea et al., 2022). We use the Jordan snow layering option, which allows for up to 100 snow layers, the UEB-2 stream approximation for canopy radiation interactions, the "Louisinv" option for stability correction of turbulent fluxes, and the "stickySnow" canopy snow interception scheme. SUMMA solves for a vegetation energy balance and canopy air temperature which is used to produce $LW\downarrow$ incident onto the sub-canopy snowpack. Additional specific modeling parameterizations with relevance to this study are described in Sec. 2.5 and the Supplemental Material. We assign a measurement height of 12m for the model forcings into the SUMMA in order to intercompare the effects of 10m tall needleleaf vegetation characteristic of the forest site at CSSL using the same forcing variables.

We use hourly data from the fifth-generation ECMWF (ERA5) reanalysis (Hersbach et al., 2020) as SUMMA model forcings after adjusting to the CSSL site. Daily precipitation from the NRCS SNOTEL site is used directly as forcing when available. Hourly ERA5 two-meter air temperatures are bias corrected to match the monthly average T_{air} observed at the CSSL from the NRCS records by removing a constant linear bias term. Comparing ERA5 $SW\downarrow$ data to observations from the CNR4 shows good agreement except for morning and evening when the surrounding vegetation blocks direct beam solar radiation. We compute and apply a $SW\downarrow$ correction coefficient as a function of the solar zenith angle to adjust the ERA5 data to account for shadows. Humidity, wind, and $LW\downarrow$ variables validated well against observations and are not adjusted.

2.4 Atmospheric Radiation Estimations and the Impacts of Clouds

High-level cirrus clouds were observed by the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite imagery at various times during the March 2026 heatwave. To model the effects of cirrus clouds on the surface radiation balance, we use the RRTMG radiative transfer model (Iacono et al., 2008), building off work from a previous study (Rudisill et al., 2025). We configure RRTMG with one hundred equally spaced pressure levels from the surface (800hPa at CSSL) to 1 hPa. Ozone, water vapor, and thermodynamic profiles for the March 2026 heatwave are acquired from ERA5 for the grid cell encompassing CSSL.

We define a cirrus cloud radiative effect (cCRF) as the difference between R_{net} at the land surface from a cloudy atmosphere against an equivalent, hypothetical clear sky atmosphere following Ramanathan et al. (1989) and subsequent studies. To compute cCRF (Table 1), we run RRTMG twice (Sec. 2.4). First, we run the model with ERA5 thermodynamic profiles and no clouds for the month of March 2026. The cirrus cloud properties are described in the supplemental material. The purpose of this experiment it to provide a theoretical bound on potential cirrus radiative forcing during the heatwave, rather than a quantitative estimate of the integrated cCRE during the event, which would require estimating/retrieving cirrus properties and frequency throughout the duration of the event.

2.5 The Snowmelt Energy Balance

Using meteorological sign conventions for turbulent fluxes, the energy budget of a snowpack is given by:

$$\text{Melt/Refreeze} + \Delta\text{CC} = \underbrace{H_\text{s} + H_\text{l}}_{\text{Turbulent Fluxes}} + \underbrace{LW\downarrow - \sigma\epsilon T_{snow}^4}_{\text{longwave}} + \underbrace{SW\downarrow(1-\alpha)}_{\text{shortwave}} \tag{1}$$

,

where Melt/Refreeze is the rate of melt or liquid water refreezing in mass per second multiplied by the latent heat of fusion, ΔCC is the change in the snowpack cold-content, H_s and H_l are the turbulent fluxes of sensible and latent heat respectively at the snow/atmosphere interface, T_{air} is the temperature of the air, T_{snow} is the temperature of the snow surface, ϵ is the emissivity of the snow (0.98), $SW\downarrow$ is the downwelling shortwave radiation, and α is the albedo of the snow surface defined as the ratio of upwelling shortwave radiation ($SW\uparrow$) to $SW\downarrow$.

The bulk expression for the H_s implemented in SUMMA is given by:

$$H_s = c_p \rho C_s u (T_{air} - T_{snow}), \tag{2}$$

where c_p is the specific heat capacity of dry air (constant), u is the horizontal wind speed, ρ is the dry air density, and C_s is the bulk transport coefficient. In this work, we use the SUMMA formulation for C_s, which models the stability correction as a function of the bulk Richardson number.

The mass-balance of the snowpack is related to the energy balance (Eq. 1) in the following ways. The daily or monthly change in SWE (mSWE, dSWE) is related to both Melt and to a lesser extent H_l, though the latter may remove 10% to upwards of 50% of the seasonal snow mass depending on a variety of factors (Sexstone et al., 2018; Schwat et al., 2025). Melt may be either positive (melt) or negative (refreezing) using the sign convention in Eq. 1. Melt integrated throughout an entire day is not synonymous with dSWE, since meltwater may be retained in the snowpack and refreeze, and (in SUMMA) only manifests as declining SWE if meltwater percolates out of the bottom snow layer to the soil. In the SUMMA model, meltwater flow throughout the snowpack layers is modeled as a gravity-driven power law.

2.5.1 The Influence of Canopy

Vegetation canopy influences the snow energy balance through several mechanisms. First, falling snow is intercepted by the canopy and may melt or sublimate from the canopy before reaching the snowpack, so total SWE accumulation under canopy is commonly less than adjacent open areas (CITE). Second, canopy layers may inhibit turbulent exchange of mass and energy by suppressing wind. Third and with most relevance to this work, vegetation impacts the radiative balance of the snowpack by altering $SW\downarrow$ and $LW\downarrow$ impacting the snow surface.

$$LW\downarrow_{\text{under canopy}} = \underbrace{(1-\varepsilon_c)\, LW\downarrow_{\text{atm}}}_{\text{sky LW through canopy gaps}} + \underbrace{\varepsilon_c\, \sigma\, T_c^4}_{\text{canopy emission}} \tag{3}$$

In SUMMA, the emissivity of the canopy layer (ϵ_c) is purely a function of the leaf-area index (LAI) and stem-area index.

2.5.2 Defining Radiative Forcing We apply the radiative forcing concept in several ways in this study (Ramanathan et al., 1989; Deems et al., 2013; Rudisill et al., 2025). In this context, radiative forcing quantifies the instantaneous difference between the net or a single stream of radiation between observed conditions and a counterfactual reference state.

To investigate the canopy radiative forcing (RF_{canopy}), we define the same for $LW\downarrow$ and/or $SW\downarrow$ underneath the canopy, minus $LW\downarrow$ and/or $SW\downarrow$ from the above canopy layer. Lacking observations sufficient to constrain RF_{canopy}, we use the SUMMA model to evaluate this quantity.

To investigate the relative contribution of α decay on the surface radiation balance, we quantify the α radiative forcing (RF_{α}) by differencing SW_{net} computed using observed α versus SW_{net} computed using observed $SW\downarrow$ and a hypothetical fresh-snow broadband α. Following previous work we assume a fresh snow α of 0.85. All radiative effects are signed such that positive indicates energy is added to the snowpack via that process (clouds, canopy, or α decay), and a negative sign indicates the opposite.

Table 1. Definitions of radiative effects used in this study. All effects are signed positive when energy is added to the snowpack. F refers the radiative component (SW or LW).

Name	Abbreviation	Equation
Cirrus Cloud Radiative Forcing	RF_{cirrus}	$F_{\text{net, cloudy}}$ - $F_{\text{net, clear}}$
Canopy Radiative Forcing	RF_{canopy}	$F_{\text{net, under-canopy}}$ - $F_{\text{net, above-canopy}}$
Albedo Radiative Forcing	RF_{α}	$SW\downarrow[(1\text{-}\alpha_{\mathrm{obs}})$ - $(1\text{-}\alpha_{\mathrm{clean}})]$

2.6 Ranking Snow Water Equivalent and Heatwave Anomalies

To rank or contextualize SWE anomalies, we use the framework from Hatchett et al. (2022), which uses a percentile-based metric to assign snow drought categories. Drought categories are ranked as abnormally dry (D0), moderate drought (D1), severe drought (D2), extreme drought (D3), and exceptional drought (D4) for values between the 30th–20th, 20th–10th, 10th–5th, and 5th–2nd and below the 2nd percentiles, respectively. SWE percentiles are computed using the NRCS SNOTEL periods of record for each site.

Z-scores are used to rank near-surface T_{air} and snow ablation anomalies, given by

$$z = \frac{X - \mu}{\sigma}$$

. Baseline mean and standard deviations are computed using a common 1981-2020 normal periods unless otherwise stated.

3 Results

3.1 The March 2026 heatwave

The March 2026 heatwave resulted from an extreme and persistent ridge of high pressure that gained strength throughout March (Fig. 2a.). While the water-year-to-date precipitation (1 October—28 February) was 100% of 1991-2020 normal leading up to March 2026, there was a prolonged midwinter dry spell (Hatchett et al., 2023) from 7 January to 9 February during which SWE began ablating (Fig. 2).

Immediately prior to the March 2026 heatwave, the region experienced two high-impact storm events. Between February 14th to 19th, a total of TODO: X m of snowfall (100+ mm of SWE) fell at CSSL, the third highest four-day total since records began in 1971. This snowfall event was followed by anomalously warm T_{air} and a rain-on-snow event on the 24—25 February that brought 91 mm of rainfall, leading to a net decline in SWE (Fig. 2b,c).

The beginning of March started with average to slightly above average T_{air}. As the high pressure ridge grew over the WUS, T_{air} peaked at 22.2°C on 19 March and 23.3°C on 20 March at CSSL but remained well-above normal from 8—31 March, with daily T_{air} values at the Tahoe City COOP station yielding z-scores greater than 2 σ for four days between the 17th-27th. This station was only 0.55°C below its all-time March record high (set on 31 March 2015; consistent records began in 1909). Correspondingly, the 500 mb height anomaly exceeded 200 meters across the WUS on this date, and was greater than two σ above the detrended mean for this date Lee et al. (2023). The hemispheric wave pattern on 16 March in particular is characteristic of an amplified North American Dipole pattern (Wang et al., 2014, 2015; Singh et al., 2016). The column air temperature anomaly for the month of March illustrates the depth of the heatwave, which exceeds two σ from the near-surface to the 200 mb level (not shown).

Snow melts consistently and starting with the February ROS event for both the open and forest sites, and we find good agreement between both SUMMA and observed SWE (Fig. 2c). SWE ablated at an average rate of 7.37 mm/day (7.06 mm/day) between the first of March and the 14th of March, after which it accelerated to a rate of 25.30 mm/day (18.66 mm/day) until the end of the month in the open (forest) until ablating completely in the open by the TODO: 28th and the forest site by the TODO: 24th.

The long-term and homogenized monthly USHCN HCNv2-data showed that 2026 had the highest March T_{air} observed since the beginning of the record in 1890, and was +3.6 °C above the 1981–2020 average — a full three σ above the mean across the four-station composite. The nClimGrid observationally constrained gridded product (derived from interpolating USHCN stations and other stations) likewise shows a +4–6 °C anomaly above average in some locations, compared to the +3.6°C anomaly observed at the four-station USHCN composite.

3.2 Snowmelt anomalies

SWE starting on March 1 was at a D1 or D0 ("No Drought") level across the 12 SNOTEL sites (Fig. 4a.) While SWE was nearly 0 for the two of the lowest elevation sites (473 and 809) this was not particularly anomalous for those locations. By the end of March, 9 of the 12 stations reached 0 SWE and the remaining sites were in D3 drought ("extreme drought") classification. Out of the 9 sites with 0 SWE on March 31 have a precedent in the period of record.

Following the anomalous March heatwave, SWE completely ablated at nine of the 12 stations, reaching the minimum value ever recorded on March 31th (Fig. 4) for those sites (Fig. 4b.). One of the remaining SNOTEL sites reached D4 to D5 snow drought conditions. Similarly, the SPIReS retrieved snow covered area reached its lowest ever recorded value on March 31st value during the MODIS record. (Fig. S1.)

The rate of monthly snow ablation (mSWE) was likewise anomalous (Fig. 4c.) at most sites. Typically, most SNOTEL sites in this region gain (or remain constant) rather than lose SWE throughout March (Fig. 4b), with the exception of two of the lowest elevation sites. Nine out of 12 sites, including CSSL, measuring their highest rate of monthly SWE ablation for the SNOTEL record. The lowest elevation site (473) is the only site without anomalous mSWE, in part because the starting value was so close to zero.

3.3 Model and observed snowmelt rates and energy balance in forest and open

While snow ablated in the forest three-to-four days earlier compared to the open site, the rate of ablation during the heatwave was lower in the forest site TODO: state the ablation rate. Separating the energy balance components explains some of this behavior (Fig. 5a–f and Fig. 6a–f). In the open site, SW_{net} increases from daytime values approaching 80 W·m^{-2} to values exceeding 240 W·m^{-2} by the end of the month. SW_{net} likewise increases throughout March in the forest site, but never exceeds 80 W·m^{-2} during the event. In contrast, LW_{net} is always negative for both the day and nighttime in the open, meaning that the snow is continuously cooling via longwave radiation. The day and nighttime rates are approximately the same. In the forest, LW_{net} is positive on average for the entirety of March 2026 except the 4th-7th, which also corresponds with the only period of significant overnight cold-content production. Overnight LW_{net} switches from slight negative ($\approx$ -8 W·m^{-2}) to positive or near-zero by the 16th. The highest rate of LW_{net} occurs on the 22nd, two days after the peak of the heat event, and exceeds 60 W·m^{-2} during the daytime.

3.4 Climatological comparison of the March 2026 energy balance

How does the March 2026 energy balance compare to typical Marches at CSSL? To do so we compare monthly averaged fluxes againt all March events from 2000-2026 (Fig. 8a–b) simulated for forest and open sites.

- The largest absolute change is in SW_{net} relative to the climatology for the open
- The largest absolute change is in LW_{net} relative to the climatology for the forest site
- TODO: more info describing differences

3.5 CanRE

Across the entire SUMMA record TODO: state how long this is, we find that CanRE$_{LW}$ (Eq. 3) increases as a function of T_{air} (Fig. 9a). This occurs in part because the model canopy is warmer than the ambient T_{air} during the daytime hours, but colder at nightime (Fig. 9b). Regardless, the emissivity effect yields a positive CanRE$_{LW}$ at all times of the day (Fig. 9c), even when the canopy is colder than the ambient T_{air}. Limited observations from the Suomi-NPP/VIIRS LST product for the CSSL site provide some external validation of the model canopy temperatures, which exceed T_{air} by several degrees during the peak of the heatwave (Fig. 9d). The extent to which the Suomi-NPP/VIIRS LST is indicative of canopy temperatures, versus canopy, exposed snow, and other land-types potentially mixed within the 1-km pixel remains to be seen and a full validation is beyond the scope of this paper; the true canopy temperature may be even warmer mid-day.

Taking into account both SW and LW effects, however, shows that CanRE$_{Net}$ transitions from positive to negative during the March 2026 heatwave (Fig. 9e), indicating that the positive addition of canopy $LW\downarrow$ emission is counteracted by the cooling associated with the canopy shading effect. This transition takes place approximately on the 14th and is due to a combination of both declining snow albedo and increasing solar insolation during the month associated with rising sun angles.

3.6 cCRF

Cirrus can impose a 4-5 W·m^{-2} idealized, instantaneous radiative forcing during March heatwave conditions, taking into account both short and longwave components. The cCRF$_{LW}$ effect increases as a function of both $T_{\rm air}$ and declines as a function of precipitable water vapor (PWV) (not shown) consistent with previous work. We caution that the 4-5 W·m^{-2} estimate is the maximum amount for cirrus clouds with the prescribed microphysical values, and that cirrus clouds were not present at all times during the heatwave event. Therefore, the net effect averaged over the duration of the even is undoubtedly lower. Moreover, the surface and TOA radiative forcing are not the same: cCRF$_{net,TOA}$ is more than five times stronger than surface (a median value of 26 W·m^{-2}). This implies that while the effect at the land surface is relatively modest, cirrus clouds are contributing to trapping heat in the upper atmosphere and may contribute to the duration and magnitude of the $T_{\rm air}$ anomaly.

3.7 AlbRF

The increase in $R_{\rm net}$ during the March heatwave is attributable to two factors —– increases in incident $SW\downarrow$ and declining α (Fig. **??**a—d). Incident daily-averaged $SW\downarrow$ increases during the March heatwave in conjunction with the rising sun-angle. Notably the ERA5 data corresponds well with the CSSL collected $SW\downarrow$ observations with a high-bias of 1.1 W·m^{-2} (Fig. **??**a).

Snow albedo simultaneously declines from an observed maximum of nearly 0.7 to values below 0.6 (Fig. **??**b) This represents a 33% increase in radiative forcing due to the α decline alone. Coupled with rising sun-angles and insolation, the snow α decline yields an AlbRF that surpasses 80 W·m^{-2} by the end of the month compared to 40 W·m^{-2} at the beginning of the month: an increase of 100% (Fig. **??**b) The SPIReS remote sensing product underestimates snow α somewhat with respect to both SUMMA and the CSSL observations, but likewise observes a decline in α during the March heatwave. The cause of the observed albedo decline corresponds with an increase in the effective snow optical grain size retrieved by SPIReS (Fig. **??**c).

Albedo is refreshed on April 1 following new snowfall to a value of 0.85. It is unclear if the dip in observed α from the 20-27th of March to values approaching 0.5 are correct. CSSL — have you found the same dip in snow α on these dates?

4 Discussion

4.1 Idealized scaling of energy fluxes with T_{air} and RH over melting snow

The results in Sec. TODO: SECTION —– and the relatively small contributions of H$_l$ and H$_s$ despite anomalously high temperatures — can best be explained by considering “offline” test cases assuming a fixed 0 °C melting snow surface. Several important considerations arise since snow cannot exceed this thermodynamic limit. In these conditions, turbulent exchange between the snow and an above 0 °C atmosphere will always be limited by the effects of buoyant suppression (Brutsaert, 2013).

Case in point, H_s over melting 0 °C snow increases linearly as a function of $T_{\rm air}$ with a slope determined by $w_{\rm spd}$, assuming an un-physical neutral condition without taking into account the effects of the buoyant suppression of turbulence (Fig. 11a). However, accounting for buoyant suppression decreases H$_s$ relative to the neutral assumption, so much so that for moderate $w_{\rm spd}$ values of 5 m·s^{-1}, H$_s$ actually *decreases* as a function of increasing T_{air} above 4°C. To better interrogate this relationship, (Fig. 11c) depicts the T_{air} value that maximized H_s for a given w_{spd}. A w_{spd} of 5 m·s^{-1} has a corresponding H$_s$ maximum at 4-5 °C; the stability effect will reduce H$_s$ at T_{air} values above or below this (Fig. 11c), and a w_{spd} of 2 m·s^{-1} achieves a maximum H_s when T_{air} is only a degree or two above freezing.

Evaporation and/or sublimation is likewise impacted by the 0 °C limitation — to a reasonable approximation, the vapor pressure of pooling liquid water or 0°C C ice crystals on top of the snow never exceeds 6.11 hPa. Therefore, the vapor pressure of the air must be less than 6.11 hPa for sublimation or evaporation to occur. Assuming a constant RH of 20% (roughly the minimum observed during the March heatwave), H$_l$ decreases rather than increases as a function of T_{air} over a 0°C snow regardless of w_{spd} (Fig. 11b,d). T_{air} greater than approximately 24-25°C at 20% RH *adds energy to the snow* via condensation/deposition. The effects of thermodynamic stability further reduce H$_l$ as a function of T_{air}. In reality, though, RH is not independent of T_{air}, and generally decreases as T_{air} rises during the daytime due to water vapor conservation and dry-air entrainment from the upper levels of the boundary layer TODO: CITE. Sublimation is permitted across the full range of RH values as T_{air} approaches 0°C. At 25°C (nearly the maximum observed during the March heatwave), RH values must be less than 20% for sublimation to occur.

Assuming an emissivity value of 0.98 (Huang et al., 2016) snow emits outgoing $LW \uparrow$ at a maximum rate of 309 W·m^{-2}. The effective emissivity of the clear-sky atmosphere, as well as $LW \downarrow$, can be easily visualized by considering the Brutsaert (1975) formula which is an approximation of the full radiative transfer equations implemented in RRTMG and other models (Sec 2.4). In addition to scaling to the fourth power, small variations in RH have a large impact on $LW \downarrow$. Over melting snow, a T_{air} of 20°C and a low RH of 20% is unable to overcome the counteracting emission by the snowpack, and LW_{net} is negative on the order of 20 W·m^{-2}. For the same T_{air}, an RH of 60% more than doubles the magnitude of LW_{net} and yields a net-heating of the snow. Based on these considerations, snow is highly sensitive to variations in both temperature and also atmospheric humidity.

Taken together, increasing RH and T_{air} will tend to increase LW_{net} and also *decrease* cooling via the sublimation/evaporation mechanism. While LW_{net} may not always be a heat-source, but rather a heat-sink, mechanism for snowmelt rates, LW_{net} over melting snow increases non-linearly and monotonically as a function of rising T_{air} and is completely insensitive to wind motions LW_{net} is perhaps best thought of as "handbrake" on the rate of snowmelt; rising T_{air} during heatwaves means that this brake is applied less and less. The same is not true for H$_s$ which only scales linearly with T_{air} at very high w_{spd} values due to the counteracting effects of buoyancy.

We caution that this is only one possible parameterization for describing the effects of surface layer stability on turbulent exchange, and the assumptions of the theory (while employed in virtually all land, snow, and atmospheric models), may not hold for complex mountain terrain. Terrain mesoscale circulations, in which case turbulence isn't generated by shear between the snow surface and atmosphere, but may be generated by other features and advected over the snow, could yield greater rates of H$_s$ or H$_l$ for given meteorlogical conditions.

4.2 Vegetation impedes melt during the heatwave

While we find that CanRE$_{LW}$ scales linearly with rising T_{air}, meaning that the vegetation LW effect is exacerbated during heatwaves (rather than constant regardless of T_{air}), the net-effect (CanRE$_{net}$) transitioned from a net-positive (vegetation enhances energy entering the snow) to net-negative (vegetation reduces energy entering the snow). This finding matches the observational data; while snow ablated earlier in the forest site, the rate of melt was less than the open TODO: how much). The earlier ablation was likely explained by canopy interception, since there was less SWE at the beginning of March in the forest site compared to the open. The ablation rate is explained by both the declining snow albedo increasing the sensitivity to $SW\downarrow$ and rising sun-angles throughout March. In other words, the SW-shading effect of the canopy is relatively minimal when the snow is bright and the strength of the sun is weak and fails to outweigh the added LW effect.

Given that this heatwave occurred early in the season compared to those observed in TODO: cite Rhoades, we hypothesize that same analysis of CanRE$_{net}$ would likely demonstrate a stronger cooling effect during later-season heatwaves with stronger solar insolation. Therefore, we conclude that *moderate canopy cover yields snowpacks that are more resilient to heatwaves than equivalent open areas* given that solar radiation is still such a strong snowmelt forcing during such conditions. This finding is consistent with reduced snow duration following the Caldor fire in the American river basin, immediately to the south of our study region (Cowherd et al., 2026). (Hatchett et al., 2023) likewise found that mid-winter dryspells increase snowmelt rates in post-fire environments in the Sierra nevada due to increased solar insolation from canopy loss by wildfire.

4.3 Elevation, slope, and aspect

We only consider flat, non sloping surfaces in this analysis (with a by-definition undefined aspect). In reality the slope and terrain aspect have a substantial influence on the $SW\downarrow$ incident on the snowpack, as well as $LW\downarrow$ from surrounding terrain.

4.4 The role of antecedent cold-content

- The amount of energy required to overcome the cold content is far less than what is required to melt ice.

4.5 Heatwaves in the future and implications for water management

- CMIP6 models underestimate monthly high temperature extremes by 11–12% (Duan et al., 2025).

- Statistical WSFS models may overestimate runoff during anomalously warm periods Lehner et al. (2017b) Lehner et al. (2017a) if historical "runoff ratios" (the amount of streamflow relative to precipitation) are assumed in the model. We have shown that such reductions in runoff ratio are more likely to be cause by soil evapotranspiration rather than snow sublimation.

- This heatwave happened before April 1, which is the date that is historically used for water resource management decisions. Post April heatwaves may be more challenging for water management under current practices — this happened in the Spring of 2022 TODO: CITE DWR Interview. Once allocations have been made by April, early snowmelt could prove costly for subsequent decisions informed by expected allocations such as planting acreage or types of crops that now may not have water available for later season irrigation. Similarly, municipal water agencies may not receive expected water late in the season. Fish, recreation, same problems? Basically, snoweaters mess everyone up later on!

- "Slower slowmelt in a warmer world" concept (Musselman et al., 2017) probably still holds but merits additional investigation. E.g., 2023 had rates on the order of 3 in/day in July compared to the 1-1.5 in/day during the March heatwave of 2026. The point is that the max rate of snowmelt for the entire water year may decline, but early season heatwaves are already increasing the melt rates for a given month or day of year.
- TODO: Discuss α decline mechanisms — how does liquid water impact α and how is this T_{air} dependent

5 Conclusions

In this study, we used energy balance observations from CSSL, NRCS Snotel observations, energy balance modeling (SUMMA), atmospheric reanalyses (ERA5) and remote sensing products (SPIRES; VIIRS/Suomi-NPP LST) to investigate interactions between a three-sigma heatwave and snow ablation that occured during March 2026 in the Northern Sierra Nevada mountains. Following an anomalously warm snow year with average to near average precipitation, three-quarters of the SNOTEL sites in the study region experienced record low end of March SWE following record rates of snowmelt during the March heatwave (Fig. 4).

- dSWE during the heatwave was relatively insensitive to initial snow cold-content conditions TODO: add figure that shows this.
- Cirrus clouds have an instantaneous 4-5 W·m^{-2} cCRF$_{Net}$ at the surface. The effect is greater at the top-of-atmosphere and likely contributes to the heatwave strengthening.
- The snow α decline driven by the natural grain-growth feedback during the heatwave lead to a large increase in SW_{net}— AlbRF went up by $\approx$ 60 W·m^{-2}
- Even though it's really hot, sensible heat H_s has a relatively small role unless it's really windy due to the effects of buoyancy.
- Increasing T_{air} increases snow to atmosphere temperature gradients but also inhibits turbulent exchange due to the buoyancy/stability effect.
- Sublimation did not have a major role in the heatwave TODO: how much sublimated relative to melt, in %. Idealized thermodynamic arguments show that sublimation is limited to very dry airmasses at high T_{air} values. Evapo-sublimation does not scale linearly with warming during heatwaves; in fact, there was less sublimation during the March heatwave than in a typical March.
- SW_{net} is still a dominant snowmelt forcing term for open areas during the heatwave — the forest site, however, is LW_{net} dominated.
- Forest canopy added resiliency to snow water retention against the heatwave —– CanRE$_{LW}$ increases with T_{air}, but this effect was not strong enough to overcome the net-cooling effect from canopy shading. Both the model and observations agree that ablation rates were lower under the canopy (-18.66 mm/day) than in the open (-25.30 mm/day) during the peak of the heatwave.

- Future work on snow/heatwave interactions should focus on the co-variability with w_{spd} and humidity with T_{air} extremes. Turbulent fluxes strongly influenced by w_{spd}, and RH influences both H_l, clear-sky $LW\downarrow$, and the cCRE.

6 Data Availability

ERA5 atmospheric data and near-surface meteorological forcings for SUMMA were downloaded from the Copernicus data store ECMWF (2023). Geopotential height anomalies and standard deviations were precomputed and provided from the supplemental material of Lee et al. (2023). SPIReS near real time data are publicly available from the National Snow and Ice Data center Snow Today page. The SUMMA model is publicly available on GitHub. The VIIRS/Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature was processed and downloaded from Google Earth Engine TODO: CITE GEE.

A Stability Correction functions in SUMMA

We use the "Loisinv" stability correction function implemented in the SUMMA model and assume that $z_h = z_0$

$$C_h = \frac{k^2}{\ln(z/z_0)^2} \cdot f(\mathrm{Ri}) \tag{4}$$

Where Ri is the bulk Richardson number, defined as

$$\mathrm{Ri} = \frac{9.81(T_{air} - T_{snow})}{0.5(T_{air} + T_{snow})u^2}. \tag{5}$$

and f(Ri) is a function with range (0,1] that declines as a function of increasing Ri.

B Vegetation emissivity

TODO: Show how SUMMA models emissivity e_c as a function of LAI and SAI

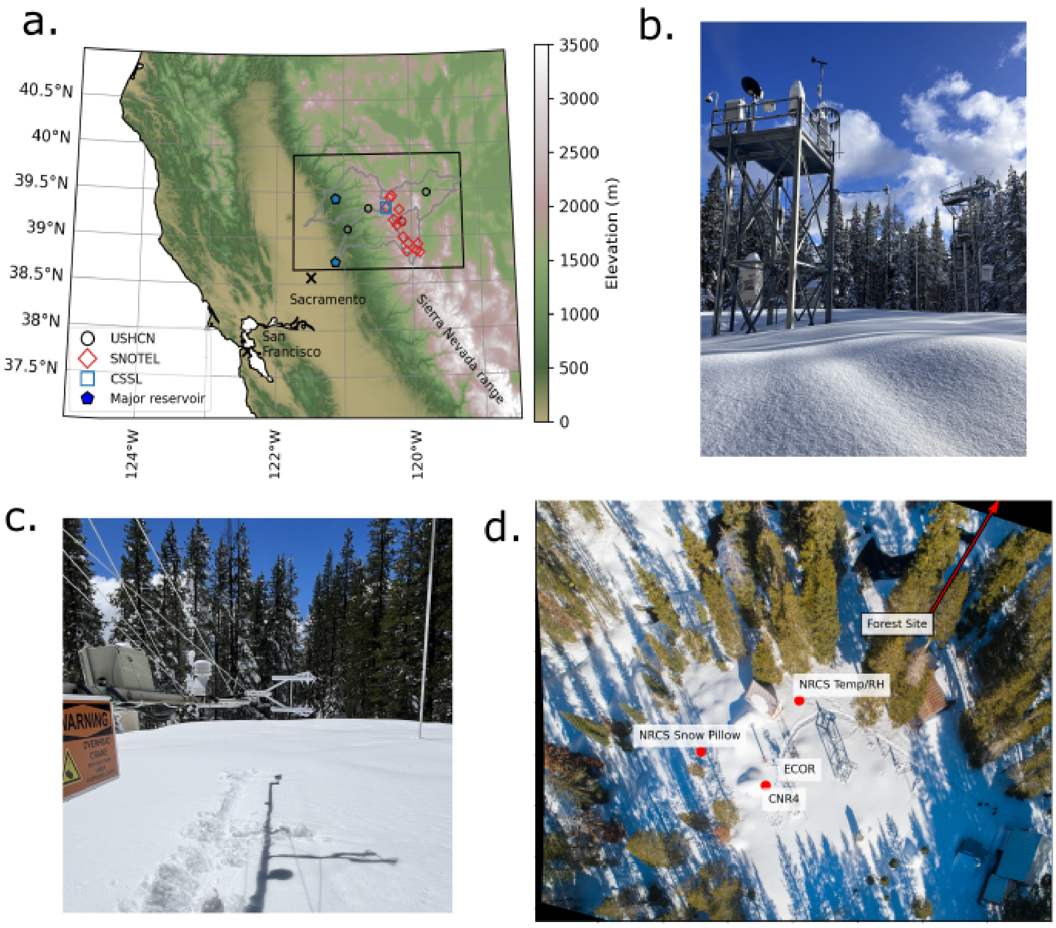

Figure 1. TODO: change the font so it's the same as the other figures**(a)** The northern California Sierra Nevada study region with terrain elevation. Outlines depict the Yuba, North Fork American, Lake Tahoe, and Truckee HUC8 watershed boundaries. Red open diamonds depict the location of NRCS SNOTEL sites, open black circles show the location of the four USHCN stations in the region, and the open blue square depicts the CSSL energy balance site. **(b)** The CSSL meteorological tower in winter, **(c)** a close-up view of the Campbell CNR4 radiometer and IRGASON eddy covariance system, and **(d)** an aerial overview of the CSSL site. The location of the forest meteorological tower (not shown) is located just outside of the frame.

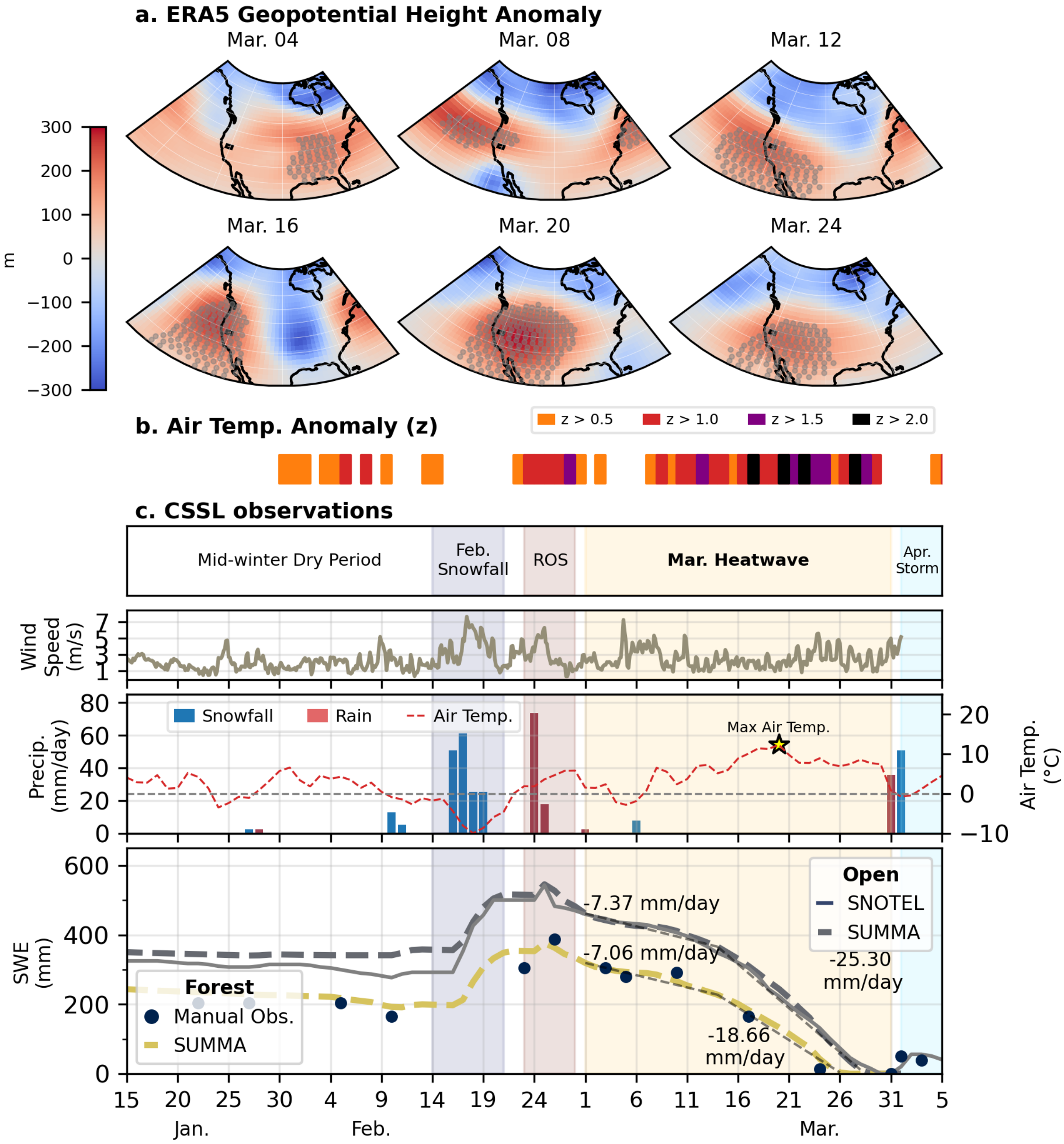

Figure 2. TODO: Change "Melt" to "Ablation". Change mar to March and Feb to February The meteorological conditions leading up to and during the March 2026 heatwave. **(a)** The 500 mb geopotential height anomaly during the March 2026 heatwave from ERA5 for six days across the WUS. The study region is outlined using an overlain black rectangle. Gray stippling indicates regions with 500 mb geopotential height anomalies greater than 2-σ above the 1950-2022 mean for the month of March. **(b-c)** shows the time series of meteorological conditions at CSSL before and during the heatwave event. **(b)** depicts the daily average two-meter air temperature and daily precipitation rate. Days with a majority of rain (red) and a majority of snowfall (blue) are differentiated. **(c)** depicts the daily SWE and the one-day change in SWE (dSWE; right axis). Days with a positive dSWE (blue) are denoted by accumulation, and days with a negative dSWE (red) are termed melt days, acknowledging that some SWE may be lost by other processes.

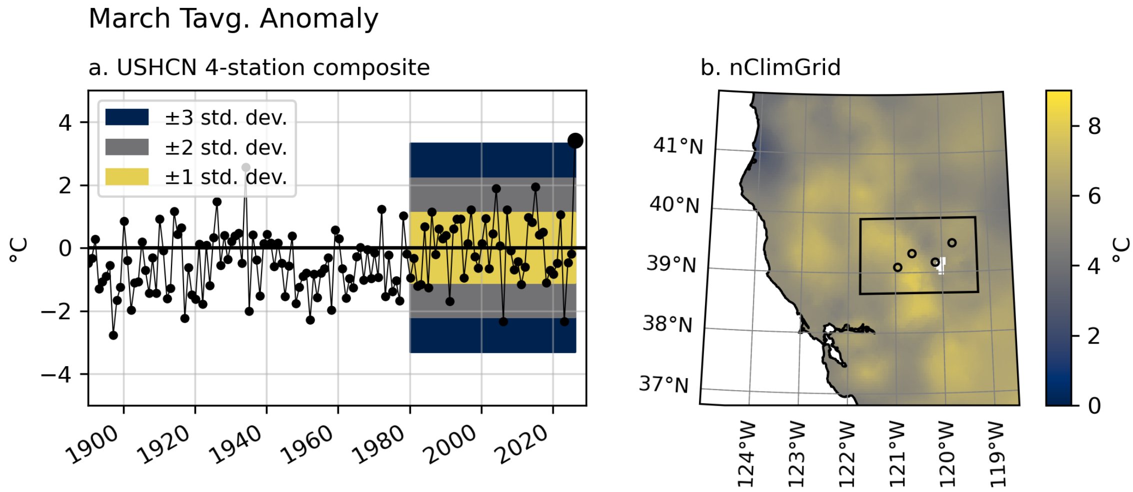

Figure 3. (a) Composite T_{air} anomalies from the four HCNv2 dataset stations in the greater Tahoe region study area for the month of March, from 1890 to 2026. Deviations from the 1980–2026 mean are depicted for one, two, and three standard deviation regions (shading). Anomalies are computed with respect to the 1980–2026 mean T_{air}. (b) The same, but for a map-view of the T_{air} anomaly of the gridded nClimGrid product.

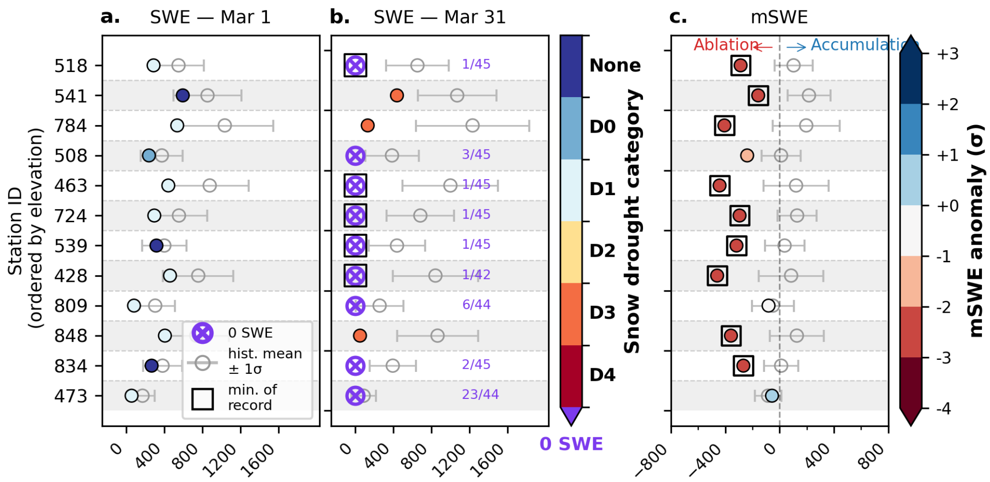

Figure 4. (a) SWE values on March 1st the 12 SNOTEL sites in the study domain, labeled by their three-digit station code. (b) The same as (a), but for SWE on March 31. Color values correspond with the snow drought D-scale. 0 SWE values are depicted by purple circle markers and the purple text lists how common 0 SWE was observed on that date for that site out of the period of record. (c) The SWE change between Mar 31 and Mar 1 compared with the historical record. Color values depict the deviation from the mean in units of standard deviation. Square markers bounding symbols indicates the SWE or dSWE is the record minimum for the period of record.

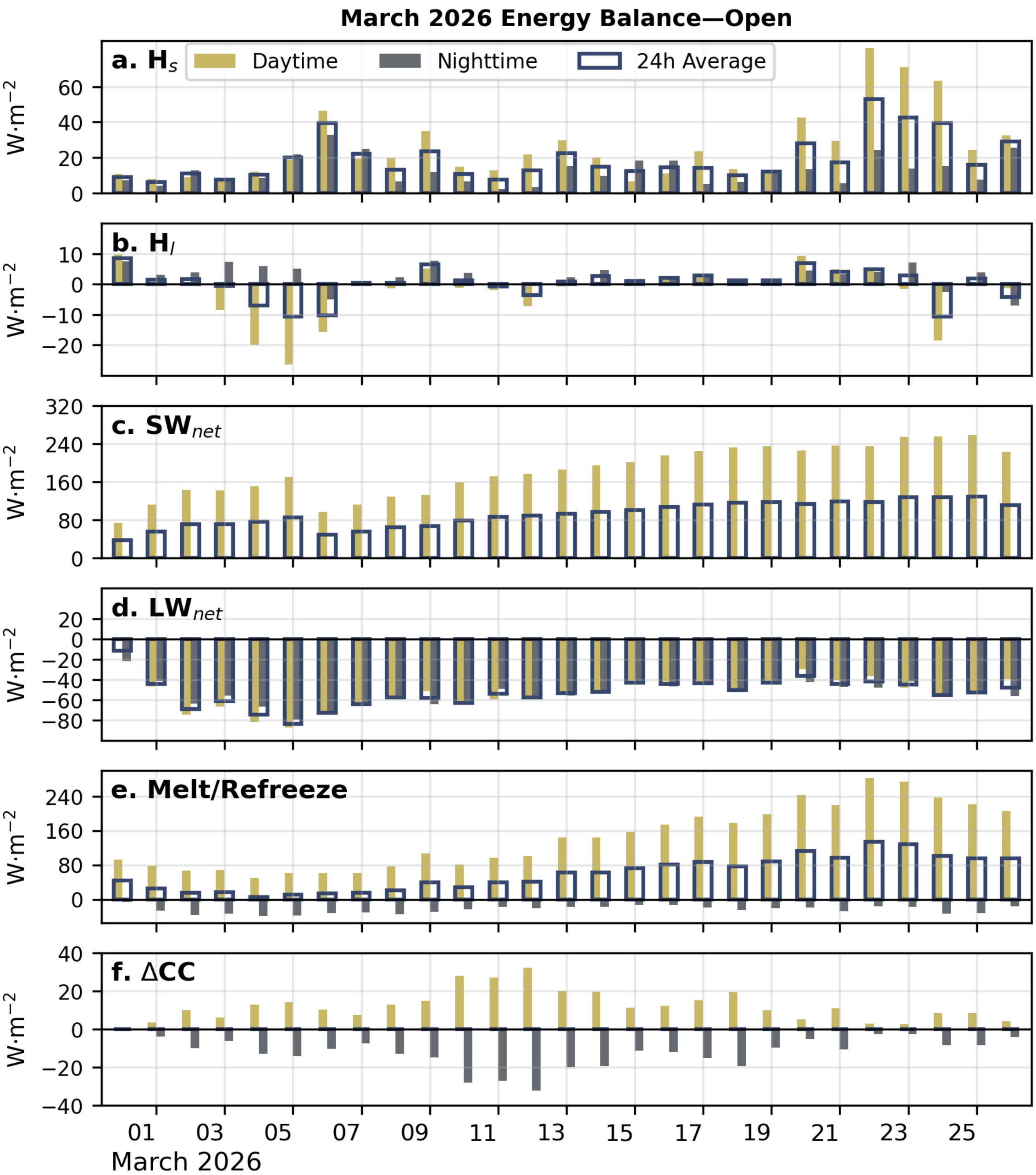

Figure 5. Daily average energy flux components including (a) H_s, (b) H_l, (c) SW_{net}, (d) LW_{net}, (e) snowpack Melt/Refreeze and (f) change in cold-content (Δ CC) during the March 2026 heatwave the open site. Each day has three bars: the left-most yellow bar depicts the daytime flux, the right-most gray bar depicts the nighttime flux, and the outline spanning both bars depicts the 24-hour average flux. Please note that the y-scales are not equivalent across all of the figures.

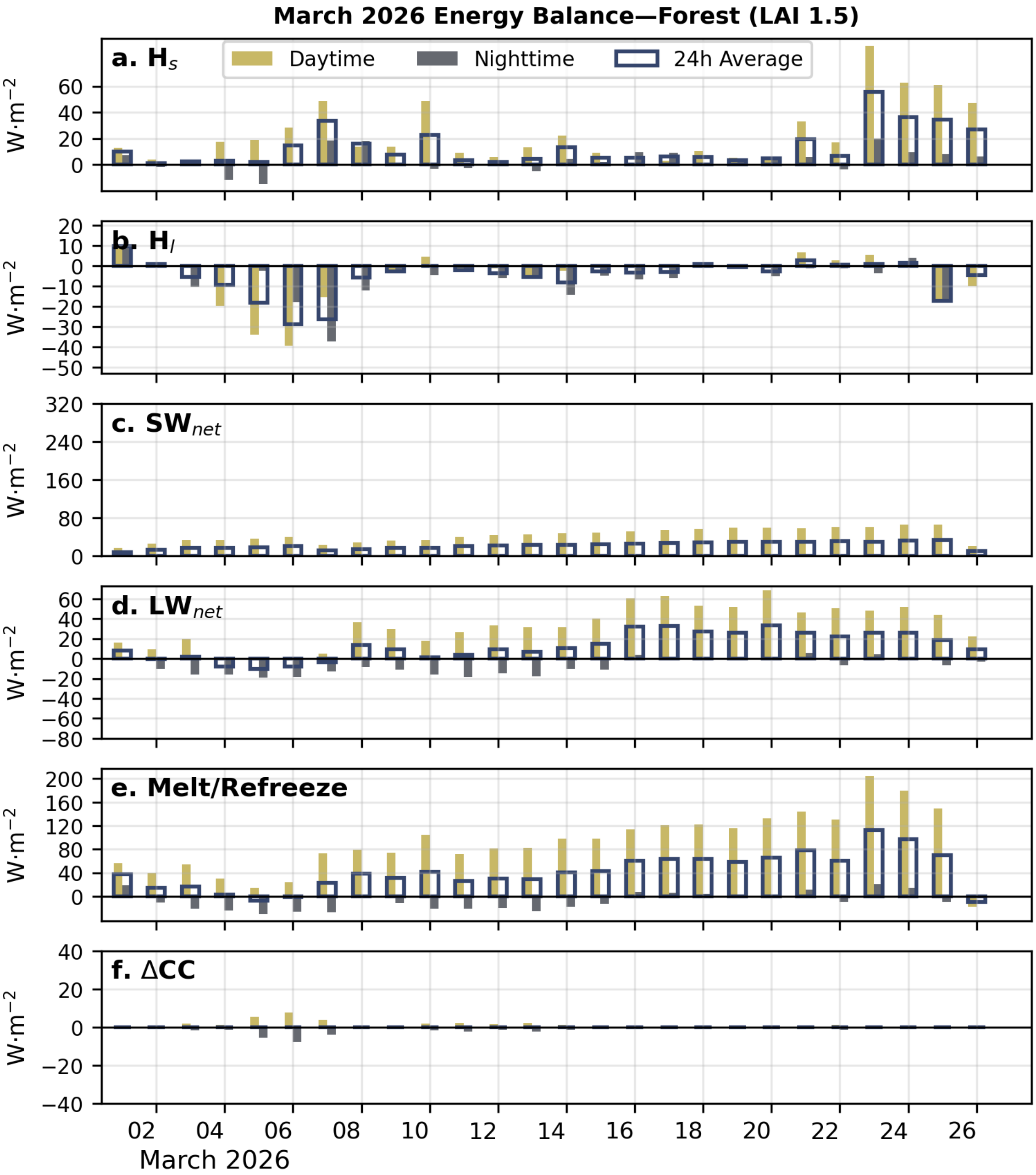

Figure 6. The same as Fig. 5 but for the forest site. Note the y-axis limits and ticks are not the same as (Fig. 5)

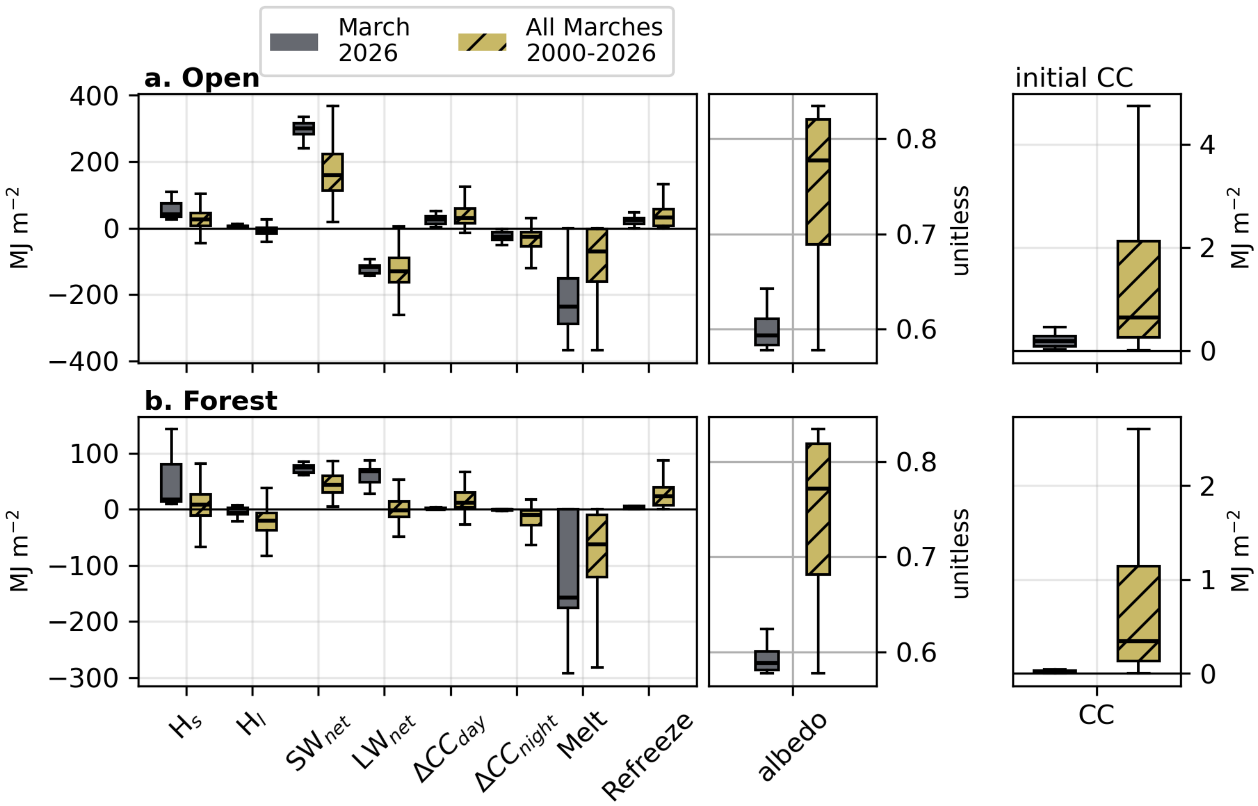

Figure 7. A comparison of daily average energy balance fluxes in the (a) open and (b) forest sites for March 2026 versus all March periods from 2000–2026. Energy balance fluxes include H_s, H_l, $SW_{\rm net}$, $LW_{\rm net}$, Melt/Refreeze, and Δ CC. Only timesteps where SWE > 0mm are included in the analysis. Box-and-whisker plots show the mean, interquartile range, and 5th and 95th percentile values.

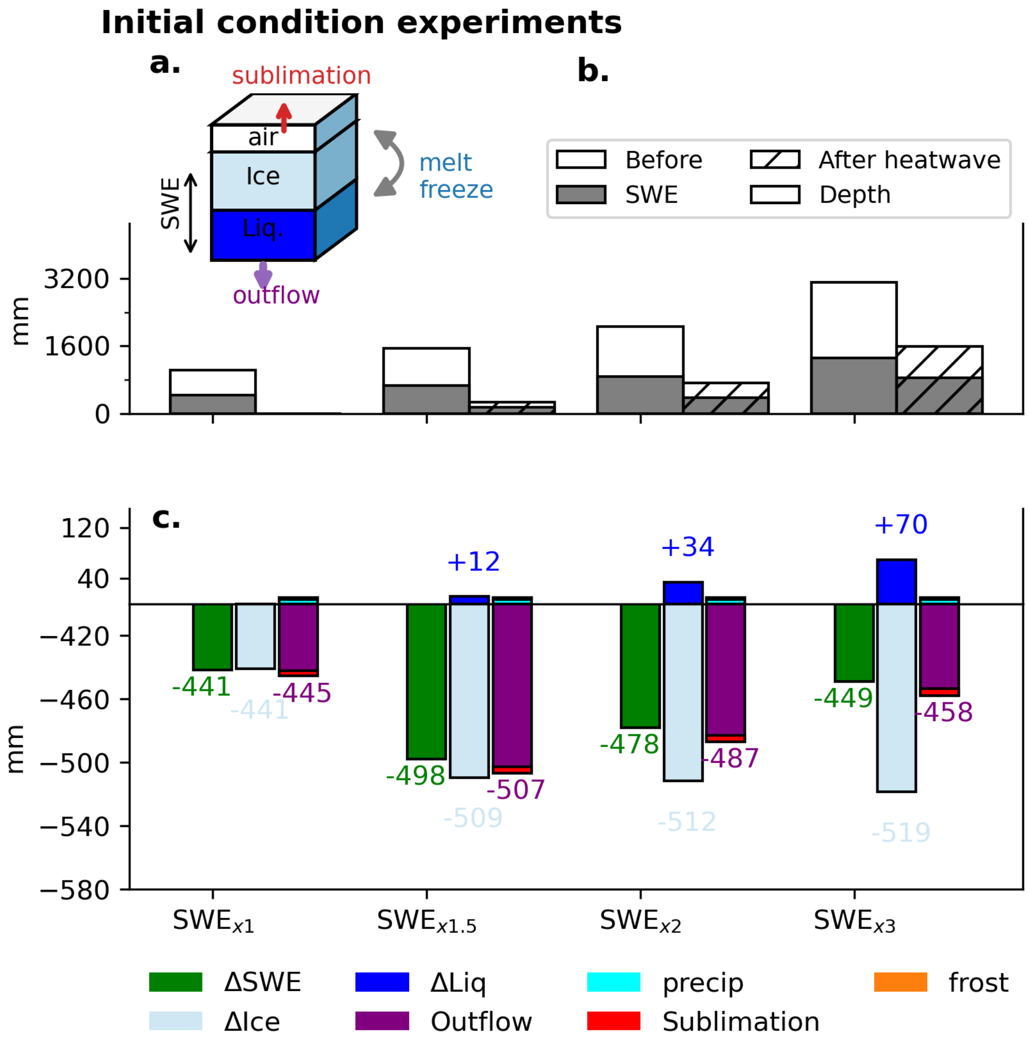

Figure 8. Initial condition tests

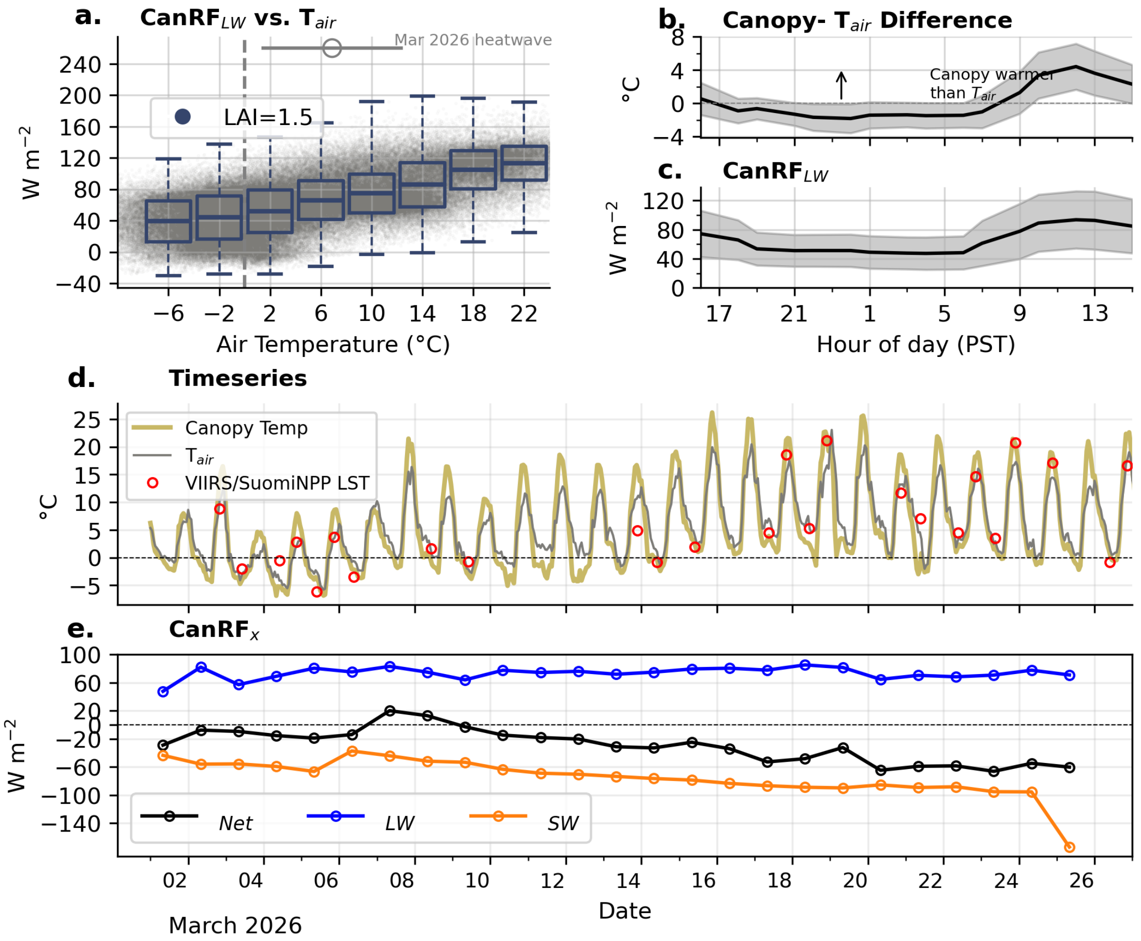

Figure 9. Examining the behavior of the canopy longwave radiative effect (CanRE): (a) CanRE vs. $T_{\rm air}$ for the entirety of the SUMMA model record (2000—2026), (b) The difference between the canopy temperature and $T_{\rm air}$ as a function of time-of-day in local time (c) the same as (b), but for CanRE, (d) a timeseries of $T_{\rm air}$, canopy temperature, and the VIIRS/Suomi-NPP LST product for the CSSL site during the month of March, and (e) a timeseries of CanRF$_{LW}$, CanRF$_{SW}$, and CanRF$_{Net}$ at the CSSL forest site during the March heatwave.

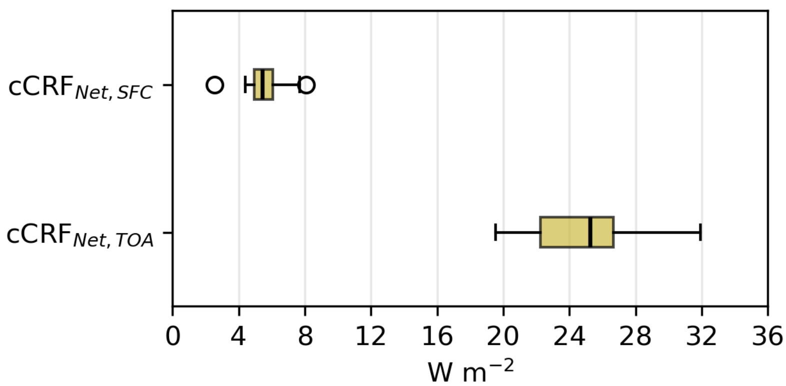

Figure 10. The daily-averaged cCRF$_{Net}$ during March 2026 as simulated by RRTMG for the surface (cCRF$_{Net,sfc}$) and top-of-atmosphere (cCRF$_{Net,TOA}$).

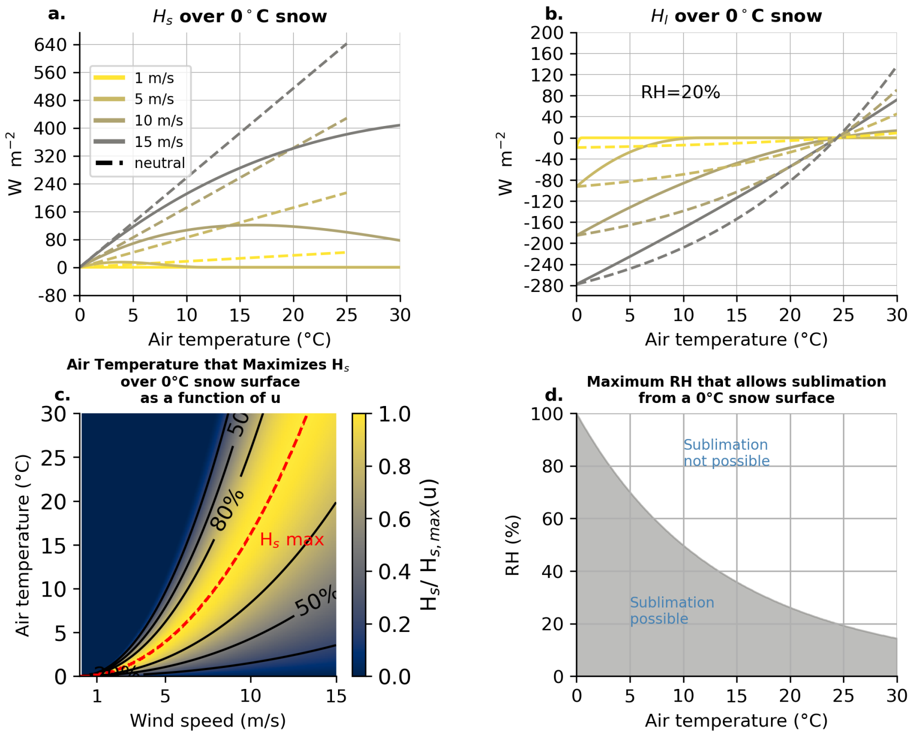

Figure 11. (a) Idealized sensible heat fluxes (H_s) into a melting snowpack as a function of T_{air} for four wind speeds (1, 5, 10, and 15 m s^{-1}) at a 12 m height for open-conditions and a snow surface roughness of z_0=0.002 m. Dashed lines show neutral-stability bulk transfer; solid lines show the same formulation with a stability correction applied as implemented in SUMMA. (b) the same as (a), but for H_l with a fixed RH of 25%. (c) The T_{air} that yields the maximum H_s into a 0 °C snowpack after correcting for stability effects (dashed red line) for a given wind speed (u). Contours depict the fraction of the maximum H_s for that wind speed value. (d) The thermodynamic limitation of sublimation as a function of T_{air}, depicting the maximum RH that permits sublimation from a 0 °C snowpack.

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